SOMAIYA
VIDYAVIHAR UNIVERSITY

06·06·2023 (E)

Maximum Marks: 100	Semester: January 2023 to May - 2023 Examination: ESE Examination - KT		Duration:3 Hrs.
Programme code: 1 Programme: B. Tech. Computer Engineering		Class: T.Y.	Semester: V (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computer Engineering	
Course Code: 116U01C501	Name of the Course: Software Engineering		

Instructions:
1)Explain with neat and suitable diagram whenever necessary diagrams
2) All questions are compulsory
3) Assume suitable data wherever necessary

Que. No.	Question	Max. Marks
Q1	Solve **ANY FOUR**	20
i)	What is process metrics?	5
ii)	What is Static and Dynamic modelling of a system? Mention merits and limitations of one over other.	5
iii)	What is cohesion and coupling in connection with object oriented design?	5
iv)	What is validation and verification with respect to software development?	5
v)	What are the risk handling approaches?	5
vi)	What are goals of software testing ?	5
Q2 A		10
i)	**Mention assumptions and constrains of the system clearly.** For a software for online banking, write functional and non-functional requirements clearly.	5
ii)	What is Component Based Software Engineering? Mention the merits of the same. OR	5
Q2 A	After clearly mentioning assumptions, compute the function point, productivity, documentation, cost per function for the following data: 1. Number of user inputs = 24 2. Number of user outputs = 46 3. Number of inquiries = 8 4. Number of files = 4 5. Number of external interfaces = 2 6. Effort = 36.9 person month 7. Technical documents = 265 pages 8. User documents = 122 pages 9. Cost = 80000 INR / month Various processing complexity factors are: 4, 1, 0, 3, 3, 5, 4, 4, 3, 3, 2, 2, 4, 5.	10

Q 2 B	Solve ANY ONE	10
i)	You have developed a software for calculation of discount to be offered to a customer based on the following criteria: **Write assumptions, if any, clearly, before writing test cases** a) 5 % discount on the first purchase if its is more than 2000 INR b) 10 % discount for registered customers (other than first time) of the purchase value. c) Offer a gift voucher worth 500 INR if the purchase is more than 5000 INR. Develop test cases to ensure each of the above is tested. (use any approach)	10
ii)	You are appointed as a software consultant for implementing a Library Management System. One of the module is returning a book. Your system should debit overdue charges, if applicable, before updating the return information. A separate log is created of all books returned on a specific date. This will help to provide a facility to generate reports of books returned day-wise. Draw Database Diagram and sequence diagram **Mention ALL YOUR ASSUMPITIONS and CONSTRAINTS of the system developed.**	10
Q3	Solve ANY TWO	20
i)	What are different categories of risks? Explain various sources of risks in a software development.	10
ii)	What is pattern based software design? How layering approach improves software development process?	10
iii)	Compare Object Oriented Testing approach with typical approach used for software testing	10
Q4	Solve ANY TWO	20
i)	What are User Interface Design principles? What steps you will follow while designing user interface for a returning a book in a software developed for a Library Management Software. **Mention your assumptions clearly, if any.**	10
ii)	What are approaches of implementing integration testing? How stubs and drivers are used in software testing?	10
iii)	Compare and contrast the object oriented methodology of Booch, Rumbaugh, and Jacobson	10
Q5	**Solve ANY FOUR**	20
i)	What are Product metrics? Give example of the same.	5
ii)	What do you understand by requirement qualities?	5
iii)	How are the changes in Software Development managed?	5
iv)	What are the advantages of a modular software design?	5
v)	With example, differentiate between packages and interfaces.	5
vi)	What is reengineering? How will that can be implemented in producing software?	5

SOMAIYA VIDYAVIHAR UNIVERSITY

08·06·2023(E)

Maximum Marks: 100	Semester: January 2023 –May 2023 Examination: ESE Examination – KT	Duration:3 Hrs.
Programme code: 01 Programme: B.Tech Computer Engineering	Class:T Y	Semester: ~~VI~~(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer	
Course Code:116U01E514	Name of the Course: Soft Computing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Design two input OR logic using McCulloch Pitts Neuron Model	5
ii)	With mathematical functions list4 different activation functions used in neurons	5
iii)	Compare BAM and Hopfield network	5
iv)	Sketch and discuss the architecture of ART network	5
v)	Define Support, Core of fuzzy set with example	5
vi)	With an example explain Center of gravity defuzzification method	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Write a short note on Neocognitron model.	5
ii)	Explain Triangular and Trapezoidal membership function of fuzzy set	5
	OR	
Q2 A	State Hebbian Learning algorithm. Show it's implementation using suitable example	10
Q 2 B	Solve any **One**	**10**
i)	Explain with examples linearly and non-linearly separable pattern classification	10
ii)	Explain Error Back Propagation Training algorithm (EBPTA) with the help of flow chart	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Design a fuzzy controller to regulate the temperature of a domestic shower. Assume that: a) Temperature is adjusted by single mixer tap b) Flow of water is constant c) Control variables is the ratio of hot to cold-water input The design should clearly mention the descriptor used for fuzzy sets and control variables, set of rules to generate control action and defuzzification. The design should be supported by figures wherever possible.	10
ii)	Train the auto associative network for input vector [-1 1 1 1] and also test the network for the same input vector. Test the auto associative network with one missing, one mistake, two missing entries in the test vector.	10
iii)	Compare different learning rules in neural network	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	20
i)	Consider the following set of input training vectors X_1, X_2, X_3 and desired response d1, d2, d3. and $X_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix}$, d1 =-1; $X_2 = \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}$, d2 =-1; $X_3 = \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$, d3 =1 Initial weight vector $w_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0.5 \end{bmatrix}$ The learning constant c=0.1 and λ=1. Use delta learning rule to calculate the weights Use bipolar continuous activation function.	10
ii)	Draw Hopfield neural network with four output nodes. Also explain training and testing algorithm of Hopfield neural network	10
iii)	Design AND Logic function with bipolar inputs and bipolar output using perceptron training algorithm. Show the weight updates up to 2 epochs.	10

Que. No.	Question	Max. Marks
Q5	Any **four**	20
i)	Write Short notes on hand written character recognition using neural network	5
ii)	Write short notes on Kohonen's self organizing network	5
iii)	Discuss the limitation of competitive learning	5
iv)	Develop graphical representation of membership function to describe linguistic variables "cold", "warm" and "hot". The temperature ranges from 0°c to 100°c. Also show plot for "cold and warm" and "warm or hot" temperature.	5
v)	Write a short notes on components of Fuzzy Logic Controller system	5
vi)	Explain how Boltzmann machine is used in constrained optimization problem	5

SOMAIYA VIDYAVIHAR UNIVERSITY 10·06·2023 (E)

Semester: Jan 2023 – May 2023 Maximum Marks: 100 Examination: ESE Examination Duration:3 Hrs.		
Programme code: 01 Programme: B. Tech	Class: TY	Semester: V(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Department	
Course Code: 116U01C502	Name of the Course: Computer Networks	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

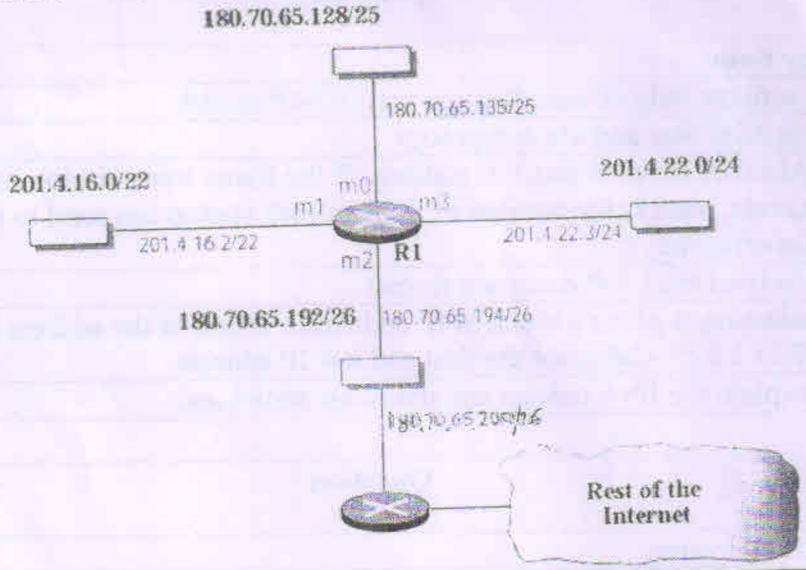
Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Explain with the help of neat diagram the TCP-IP model.	5
ii)	Explain in brief Star and Mesh topology	5
iii)	A pure ALOHA network has 100 stations. If the frame transmission time is 1 microseconds, what is the number of frame/s each station can send to achieve the maximum efficiency?	5
iv)	Explain in brief the UDP datagram format.	5
v)	An organization is given a block of IP addresses. If one of the address in the block is 172.17.15.12/23. Calculate the first and last IP address.	5
vi)	Briefly explain the IPv6 unicast and multicast addresses.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	What is Byte stuffing? Explain with the help of neat diagram.	5
ii)	The destination address in an Ethernet frame is 6A:30:10: 21:11:1A. Show how this address is sent out on the line.	5
	OR	
Q2 A	Explain with the help of neat diagram the working of Go-back-N ARQ. The efficiency of Go-back-N is superior to Selective Repeat in case of error free channel. Why?	10 (8+2)
Q 2 B	Solve any **One**	10
i)	Explain with the help of neat diagram the working of CSMA/CD MAC	10
ii)	Explain in detail the CSMA/CA MAC mechanism used in wireless LANs.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	20
i)	Explain with the help of neat diagram IPv4 header format explaining the function of each field. An IPv4 packet has arrived with the first few hexadecimal digits as shown. 0x45000028000100000101 . . . What is the size of header for this packet? How many hops can this packet travel before being dropped?	10 (8+2)
ii)	Explain in brief the ARP operation. Two hosts P and Q are on the same Ethernet network. Host P with IP address 192.168.15.21/24 and physical address B2:34:55:10:22:10 has a packet to send to	10 (4+6)

	another host Q with IP address 192.168.15.30/24 and physical address A4:6E: F4:59:83:AB (which is unknown to host P). Show with neat diagram, the ARP request and reply packets exchanged between them.	
iii)	What are different categories of ICMP messages? Explain in detail various Query-Response messages of ICMP.	10 (2+8)

Que. No.	Question	Max. Marks
		20
Q4	Solve any **Two**	10
i)	Enlist different forwarding techniques. The topology of a network is as shown below. Show the routing table of Router R1 and the forwarding process if packet arrives at R1 with destination address 180.70.65.140. (Note: Address in Bold is network address)	(2+8)



ii)	Explain in detail the working of Link state routing algorithm with the aid of neat diagrams.	10
iii)	Explain with the help of neat diagrams working of TCP connection establishment, data transfer and connection termination.	10

Que. No.	Question	Max. Marks
		20
Q5	(Write notes / Short question type) on any **four**	5
i)	DHCP	5
ii)	FTP	5
iii)	HTTP	5
iv)	Telnet	5
v)	Congestion control mechanisms	5
vi)	NAT	

13-6-2023 (E)

SOMAIYA VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: January 2023 –May 2023 Examination: ESE Examination(KT)		Duration:3 Hrs.
Programme code: 01 Programme: B. Tech in Computer Engineering	Class: TY	Semester: V(SVU 2020)	
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: COMP		
Course Code: 116U01C503	Name of the Course: Operating System		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	With respect to System Software, Describe assembler and loaders.	5
ii)	Differentiate between Compiler and Interpreter	5
iii)	Illustrate the Process State Transition Diagram with suitable diagram.	5
iv)	Explain the concept of Resource pre-emption for Recovery from Deadlock	5
v)	With respect to Linux Operating System, Define Inodes with suitable diagram.	5
vi)	With respect to Memory Management, Describe Internal and External Fragmentation.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Describe the System Boot Process.	5
ii)	Define shell. Further Comment on the different types of shells.	5
	OR	
Q2 A	Illustrate the following types of Operating System Structures with the help of suitable examples:- i. Traditional UNIX System Structure ii. Layered Approach iii. Microkernel System Structure	10
Q 2 B	Solve any **One**	**10**
i)	Compare and Contrast between Multilevel Queue and Multilevel Feedback Queue Scheduling Algorithms with the help of suitable diagrams.	10
ii)	Differentiate between User level Threads and Kernel Level Threads. Further Describe the various Multithreading Models.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared Boolean variables S1 and S2 are randomly assigned. Method Used by P1 while (S1 == S2) ;	10

	Critical Section S1 = S2; Method Used by P2 while (S1 != S2) ; Critical Section S2 = not (S1); Analyse the methods and State whether Mutual Exclusion and Progress Requirement are being satisfied or not. Justify your answer.	
ii)	Discuss the Semaphore solution for Dining Philosophers Problem.	10
iii)	With respect to Process Synchronization, Examine the Bounded Buffer Problem and Readers Writer Problem.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Consider a disk queue with requests for I/O to blocks on cylinders in order 43, 33, 127, 87, 17, 99, 20. The head is initially at cylinder number 60, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. If the following Disk Scheduling algorithms are applied:- a) C-LOOK b) C-SCAN For all the algorithms, Find the order in which the requests will be serviced. Further Calculate the total head movement (in number of cylinders) incurred while servicing these requests.	10
ii)	With respect to File Management, Illustrate the various File Allocation Methods with suitable diagrams.	10
iii)	Discuss the Address Translation Scheme to map Pages into frames with the help of suitable diagrams. Further Explain the concept of Segmentation with an example.	10

Que. No.	Question	Max. Marks
Q5	Solve any **four**	**20**
i)	Describe System Calls ,Further list the types of system calls.	5
ii)	Describe Linux Scheduling.	5
iii)	Distinguish between Message Passing and Shared Memory	5
iv)	Explain the concept of Monitors with the help of suitable diagrams.	5
v)	Discuss the various Input/Output Buffering schemes with suitable diagrams for each.	5
vi)	Discuss the structure of Hashed Page Table with the help of suitable diagrams.	5

15.6.2022(E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester: August 2022 – December 2022 Examination: ESE Examination (KT)		Duration:3 Hrs.
Programme code: 75 Programme: Minor Programme in Computer Engineering		Class: TY	Semester: V(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Comp	
Course Code: 116m75C501	Name of the Course: Operating System		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	With respect to System Software, Describe linkers and loaders.	5
ii)	Describe the System Boot Process.	5
iii)	Illustrate the Process State Transition Diagram with suitable diagram.	5
iv)	How can Semaphores be utilized for Process synchronization?	5
v)	Define Bit Vectors and Free Block List with respect to Free Space management.	5
vi)	With respect to Memory Management, Describe Internal and External Fragmentation.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	Differentiate between Symmetric and Asymmetric Multiprocessors.	5
ii)	Define shells. Further Comment on the different types of shells.	5
	OR	
Q2 A	Illustrate the different types of Operating System Structures with the help of suitable examples.	10
Q 2 B	Solve any **One**	10
i)	Describe the working of fork() and exec() system call for Process Creation. Further Discuss the different scenarios of Process Termination.	10
ii)	Compare and Contrast between Multilevel Queue and Multilevel Feedback Queue Scheduling Algorithms with the help of suitable diagrams.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	20
i)	Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared boolean variables S1 and S2 are randomly assigned. Method Used by P1 while (S1 == S2) ; Critical Section S1 = S2; Method Used by P2	10

	while (S1 != S2) ; Critical Section S2 = not (S1); Analyse the methods and State whether Mutual Exclusion and Progress Requirement are being satisfied or not. Justify your answer.	
ii)	Discuss how Mutual Exclusion can be implemented with Test & Set () and Swap() Instructions.	10
iii)	An operating system uses the banker's algorithm for deadlock avoidance when managing the allocation of three resource types X, Y and Z to three processes P0, P1 and P2. The table given below presents the current system state.	10

	Allocation			Max		
	X	Y	Z	X	Y	Z
P0	0	0	1	8	4	3
P1	3	2	0	6	2	0
P2	2	1	1	3	3	3

There are 3 units of type X, 2 units of type Y and 2 units of type Z still available. The system is currently in safe state. Consider the request for additional resources by P1. P1 requests 2 units of X, 0 units of Y and 0 units of Z. Determine whether the request by P1 should be granted or rejected.

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Consider a disk queue with requests for I/O to blocks on cylinders in order 43, 33, 127, 87, 17, 99, 20. The head is initially at cylinder number 60, moving towards larger cylinder numbers on its servicing pass. The cylinders are numbered from 0 to 199. If the following Disk Scheduling algorithms are applied:- a) C-LOOK b) C-SCAN c) SSTF(Shortest Seek Time First) d) SCAN e) FCFS For all the algorithms, Find the order in which the requests will be serviced. Further Calculate the total head movement (in number of cylinders) incurred while servicing these requests.	10
ii)	With respect to File Management, Illustrate the various File Allocation Methods with suitable diagrams.	10
iii)	With respect to Memory Management, Illustrate the Hierarchical and Hashed Page Table Structure with Suitable diagrams.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	Distinguish between Unix and Linux.	5
ii)	Differentiate between User Level Thread and Kernel Level Thread.	5

iii)	Discuss the various approaches utilized for recovery from deadlock.	5
iv)	Discuss Process Synchronization. Further , Comment on Buffering needed for Inter Process Communication.	5
v)	Describe the Inodes Control Structure utilized for File management in Unix.	5
vi)	Explain how Page table can be implemented with Translation lookaside Buffer.	5

15.6.2023 (E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	**Semester: January 2023 –May 2023** **Examination: ESE Examination** (KT)		**Duration:3 Hrs.**
Programme code: 55 **Programme:** Honours in Cyber Security & Forensics		**Class:** TY	**Semester: V (SVU 2020)**
Name of the Constituent College: **K. J. Somaiya College of Engineering**		**Name of the department: Computer Engineering**	
Course Code: 116h55C501	**Name of the Course: Block Chain Technology**		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	What is hash function? Discuss the properties of Hash function.	5
ii)	Explain ICO (Initial coin Offering)	5
iii)	What are the application areas of blockchain	5
iv)	Explain Sybil attack.	5
v)	Compare public and private blockchain	5
vi)	Discuss the working of digital signature.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	How Does Proof of Stake (PoS) Differ from PoW?	5
ii)	Explain Distributed Ledger Technology in blockchain.	5
	OR	
Q2 A	Explain Blockchain Double Spending. How Bitcoin handles the Double Spending Problem? Give Suitable Example to Explain.	10
Q 2 B	Solve any **One**	**10**
i)	Describe the following terms with example 1) Soft Fork 2) Hard Fork	10
ii)	Explain the types of Blockchain	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain Dapps, DAOs	10
ii)	Discuss the elements of Cryptography and elements of Game Theory used in Blockchain.	10
iii)	What is the smart contract? Why it is needed? How smart contract is processed.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Discuss the structure of Solidity and its specific data types	10
ii)	Discuss the case study of energy supply and supply chain.	10
iii)	Write a short note on 1) Markle Tree Root 2) Nonce 3) UTXO	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	Explain use of Ethereum Virtual Machine.	5
ii)	Explain the steps to setup Truffle environment on Machine for Ethereum.	5
iii)	How to deploy smart contract?	5
iv)	Explain Bitcoin protocol	5
v)	What is the mining? Explain proof of Burn with an example.	5
vi)	What is GHOST Protocol?	5